



AI-Based Virtual Try-On System & Fashion Recommendation

RARITONE SUMMER INTERNSHIP 2026

PRESENTED BY: PRANEETHA AYYANA SOMAYAJULU

ROLE: AI/ML INTERN

Table Of Contents



- Internship Overview
- Project Objectives
- Research Completed
- Datasets & Assets
- 2D Virtual Try-On Research
- Pose Detection & Body Segmentation
- 3D Avatar Generation
- Blender Exploration
- AI Workflow
- Challenges & Solutions
- Key Learnings
- Future Scope
- Project Summary

INTERNSHIP OVERVIEW

INTERNSHIP OVERVIEW

- ROLE: AI/ML INTERN
- DOMAIN: ARTIFICIAL INTELLIGENCE & COMPUTER VISION
- INTERNSHIP DURATION: SUMMER INTERNSHIP 2026
- FOCUS AREA: VIRTUAL TRY-ON TECHNOLOGIES

AREAS EXPLORED

- 2D VIRTUAL TRY-ON
- BODY SEGMENTATION
- POSE DETECTION
- AVATAR GENERATION
- HUMAN BODY RECONSTRUCTION
- AI RECOMMENDATION SYSTEMS

PROJECT OBJECTIVES

- Study modern Virtual Try-On technologies.
- Explore body segmentation techniques.
- Analyze pose estimation methods.
- Research garment fitting approaches.
- Investigate 3D avatar generation technologies.
- Understand human body reconstruction.
- Explore future AR/VR integration possibilities.
- Study AI-powered recommendation systems.

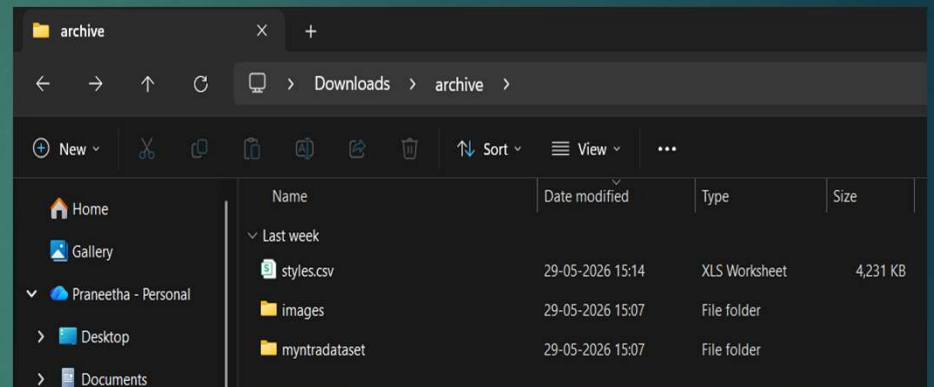
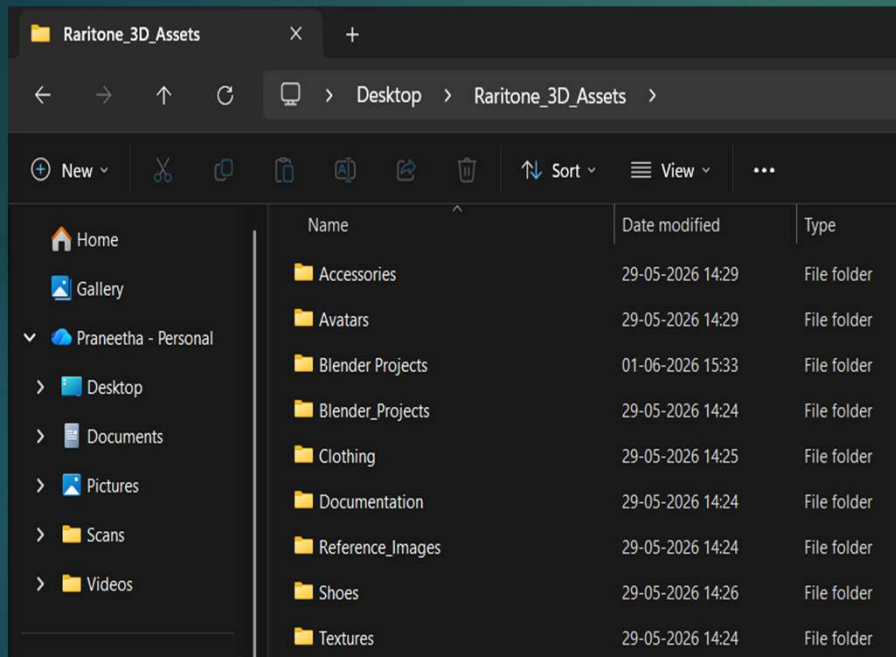
RESEARCH COMPLETED



RESEARCH AREAS COMPLETED

- 2D VIRTUAL TRY-ON SYSTEMS
- BODY SEGMENTATION
- POSE DETECTION
- GARMENT MAPPING
- CLOTH SIMULATION
- 3D AVATAR GENERATION
- HUMAN BODY RECONSTRUCTION
- AI RECOMMENDATION SYSTEMS
- AR/VR FASHION TECHNOLOGIES

DATASETS & ASSETS COLLECTED



Outcome:

Created a structured asset repository for future Virtual Try-On development and research.

2D VIRTUAL TRY-ON RESEARCH

ACTIVITIES PERFORMED

- GARMENT OVERLAY EXPERIMENTS
- CLOTHING ALIGNMENT TESTING
- POSE DETECTION ANALYSIS
- GARMENT PLACEMENT RESEARCH

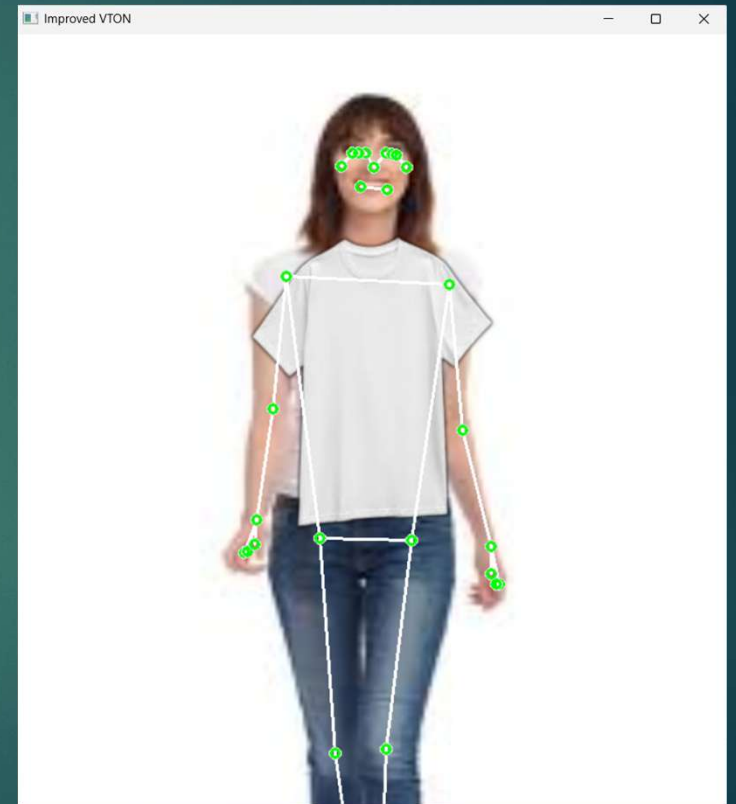
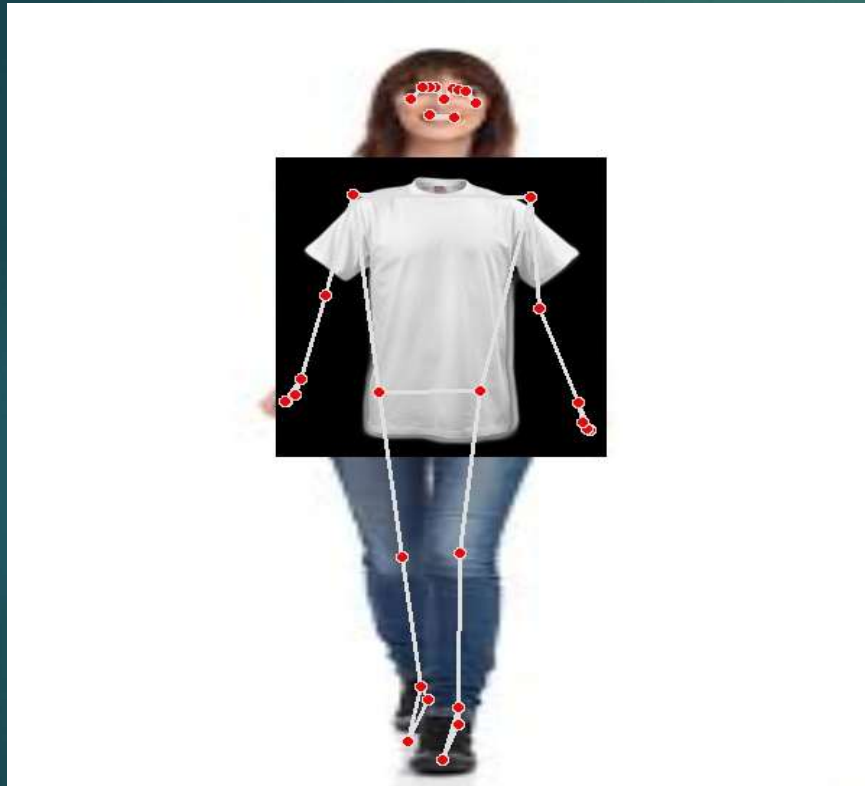
TECHNOLOGIES USED

- PYTHON
- OPENCV
- MEDIAPIPE

FINDINGS

- BODY LANDMARKS PLAY A CRITICAL ROLE IN GARMENT ALIGNMENT.
- CLOTHING PLACEMENT ACCURACY AFFECTS REALISM.
- SEGMENTATION IMPROVES OVERALL TRY-ON QUALITY.

2D VIRTUAL TRY-ON



POSE DETECTION & BODY SEGMENTATION



MODELS EXPLORED:

MEDIAPIPE

- FAST
- LIGHTWEIGHT
- REAL-TIME LANDMARK DETECTION

OPENPOSE

- ACCURATE POSE ESTIMATION
- DETAILED BODY KEYPOINTS

DETECTRON2

- ADVANCED BODY SEGMENTATION
- HIGH-QUALITY DETECTION

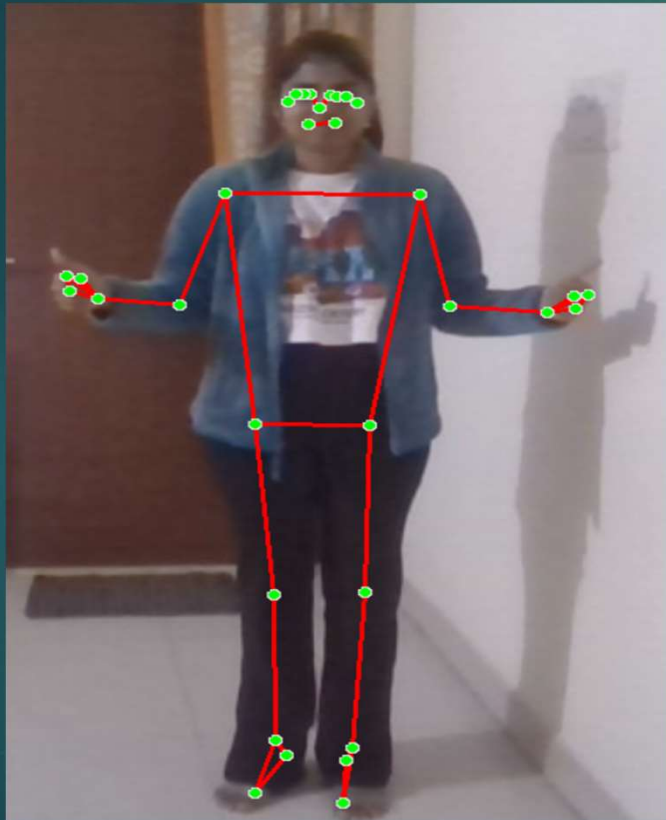
APPLICATIONS

- BODY LANDMARK DETECTION
- HUMAN POSE ESTIMATION
- GARMENT ALIGNMENT
- USER BODY ANALYSIS

Comparison model

Model	Purpose	Strength
MediaPipe	Pose Detection	Fast
OpenPose	Pose Estimation	Accurate
Detectron2	Segmentation	Detailed

POSE DETECTION & BODY SEGMENTATION



Segmentation Mask Output

The binary segmentation mask generated by the AI model.

Observation

- White region represents detected human body
- Black region represents removed background

Importance

This confirms that the AI correctly identified body regions.

Background Removed Output

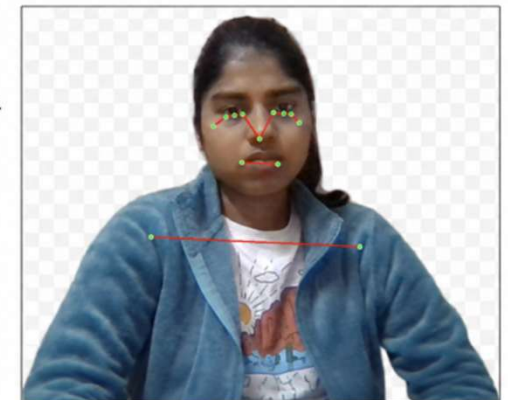
The final segmented image after removing the background.

Observation

- Human body remains visible
- Background is removed successfully

Importance

This demonstrates successful body isolation.



3D AVATAR GENERATION



TECHNOLOGIES STUDIED

- SMPL
- PIFUHD
- ICON
- HUMANNERF
- 3D GAUSSIAN SPLATTING

OBJECTIVE

TO UNDERSTAND HOW AI GENERATES REALISTIC DIGITAL HUMAN AVATARS FOR FUTURE VIRTUAL TRY-ON SYSTEMS.

MODEL COMPARISON

Technology	Purpose	Strength
SMPL	Body Modeling	Efficient & Lightweight
PIFuHD	Human Reconstruction	High Detail
ICON	Clothed Human Reconstruction	Better Clothing Detail
HumanNeRF	Neural Rendering	Realistic Appearance
Gaussian Splatting	3D Reconstruction	Fast Rendering

BLENDER EXPLORATION



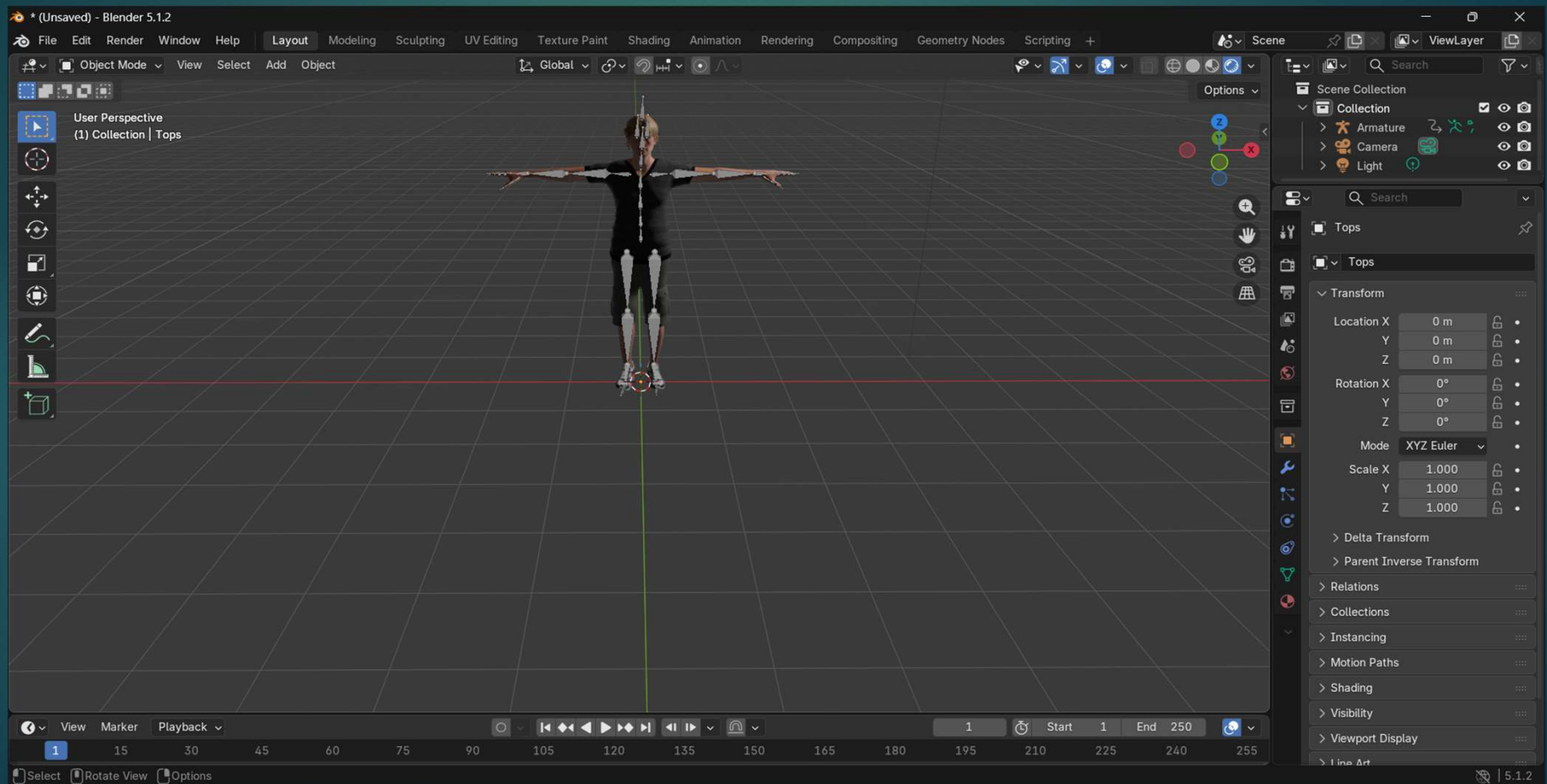
PRACTICAL ACTIVITIES

- IMPORTED AVATAR MODELS
- EXPLORED MATERIAL PREVIEW MODE
- VISUALIZED HUMAN MESHES
- INSPECTED AVATAR STRUCTURES
- PERFORMED GARMENT PLACEMENT EXPERIMENTS

LEARNING OUTCOME

GAINED PRACTICAL EXPOSURE TO AVATAR VISUALIZATION AND 3D ASSET WORKFLOWS.

BLENDER EXPLORATION



COMPLETE AI WORKFLOW

Virtual Try-On Workflow

User Image



Body Detection



Body Segmentation



Pose Detection



Feature Extraction



Garment Mapping



Avatar Generation



Virtual Try-On Result

Challenges and Solutions



CHALLENGES

- UNDERSTANDING AI FRAMEWORKS
- LEARNING BLENDER
- MANAGING DATASETS
- EXPLORING 3D RECONSTRUCTION
- HARDWARE LIMITATIONS

SOLUTIONS

- COMPARATIVE RESEARCH
- WORKFLOW-BASED LEARNING
- BLENDER EXPLORATION
- ORGANIZED DATASETS
- STRUCTURED DOCUMENTATION

IMPROVED UNDERSTANDING OF AI-POWERED VIRTUAL TRY-ON SYSTEMS.

KEY LEARNINGS



TECHNICAL SKILLS

- ARTIFICIAL INTELLIGENCE
- COMPUTER VISION
- POSE DETECTION
- BODY SEGMENTATION
- AVATAR GENERATION
- HUMAN RECONSTRUCTION

PROFESSIONAL SKILLS

- TECHNICAL RESEARCH
- DOCUMENTATION
- PROBLEM SOLVING
- INDEPENDENT LEARNING

PROJECT SUMMARY



PROJECT ACHIEVEMENTS

- STUDIED VIRTUAL TRY-ON TECHNOLOGIES
- EXPLORED BODY SEGMENTATION
- ANALYZED POSE DETECTION MODELS
- INVESTIGATED AVATAR GENERATION
- RESEARCHED HUMAN RECONSTRUCTION
- ORGANIZED DATASETS AND ASSETS

OVERALL OUTCOME

DEVELOPED A STRONG UNDERSTANDING OF AI, COMPUTER VISION, VIRTUAL FASHION, AND EMERGING TECHNOLOGIES.



Thank You

This internship provided valuable exposure to Artificial Intelligence, Computer Vision, and Virtual Try-On technologies while strengthening my research, documentation, and problem-solving skills.

Raritone Summer Internship 2026

Presented By: Praneetha Ayyana Somayajulu

Role: AI/ML Intern